Airbnb Price & Scenario Analysis Dashboard

Github Link: https://github.com/shiva3994/Projects/blob/main/Power%20BI/Capstone_Airbnb_Project%20F.pbix

User Guide — Navigation & Data Refresh: This guide explains how to navigate the report, use its interactive features, and refresh the underlying data.

1. Report Structure

The report has three main pages, accessible via the tabs at the bottom of the screen, plus two hidden supporting pages that appear automatically when needed:

Page	What It Shows
Overview	The landing page. Shows headline KPIs (Total Listings, Availability Rate, Average Price, Scenario Price, Projected Revenue, Average Host Tenure), a date timeline, category filters, the two What-If sliders, and a map of all Antwerp listings.
Listing Analysis	A closer look at the listings themselves: a larger interactive map, a price-distribution histogram, and a Top 20 Listings table ranked by review count.
Scenario Insights	Where the What-If sliders are put to work: a price comparison chart (actual vs. scenario), a month-by-month projected revenue trend, and a revenue breakdown by room type. Includes three saved bookmarks for quick scenario switching.
Listing Details (hidden)	Opens automatically when you right-click a specific listing and choose Drill Through. Shows that one listing's host, amenities, and guest reviews.
Listing Tooltip (hidden)	A small pop-up card that appears automatically when you hover over a point on the map — shows a quick price/room-type/review summary.

2. Using the Filters and Slicers

- Timeline: drag either end of the slider to narrow the whole report to a specific date range.
- Room Type / Property Type / Guests / Seasons: click any option to filter every visual on the page to just that category. Click 'Deselect all' first if you want to select only one or two specific options.
- These filters affect only the page you're currently on, except where a filter has been intentionally duplicated across multiple pages (e.g. Room Type appears on more than one page for convenience).

3. Using the What-If Sliders

Two sliders let you simulate different business scenarios without changing any real historical data:

Slider	What It Does
Occupancy Rate (What-If)	Simulates how often listings are booked, from 0% to 100%. Moving this changes the Projected Revenue figure only — it does not affect Scenario Price.
Seasonal Pricing Multiplier	Simulates a seasonal price adjustment, from 0.5x to 2.0x the current average price. Moving this changes both Scenario Price and Projected Revenue.

Try it: set Occupancy Rate to 0.75 and Seasonal Pricing Multiplier to 1.5 to see a 'busy summer' scenario. Set them to 0.30 and 0.80 to see a 'slow winter' scenario.

4. Using the Saved Bookmarks (Scenario Insights page)

Rather than moving the sliders manually, three ready-made scenarios are saved as bookmarks. Open the Bookmarks pane (View tab → Bookmarks) and click any of the following to jump straight to that scenario:

Bookmark	What It Sets
Baseline Scenario	Occupancy 0.50, Seasonal Multiplier 1.00 — the default, no adjustment.
Off-Peak Scenario	Occupancy 0.30, Seasonal Multiplier 0.80 — a slow, low-season month.
Peak Summer Scenario	Occupancy 0.75, Seasonal Multiplier 1.50 — a high-demand, high-price month.

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5. Drilling Into a Specific Listing

1. Go to the Listing Analysis page and find the Top 20 Listings table (or any visual showing individual listings).
2. Right-click on the row for the listing you want to inspect.
3. Choose 'Drill through' → 'Listing Details' from the menu that appears.
4. You'll land on a page showing that one listing's price, host, amenities, and guest reviews.
5. To return, click the back arrow in the top-left corner of the Listing Details page.

6. Refreshing the Data

If new Calendar, Listings, Hosts, or Reviews data becomes available, the report can be refreshed as below:

In Power BI Desktop

1. Open the .pbix file.
2. Go to the Home tab.
3. Click 'Refresh'. Power BI will re-run all Power Query cleaning steps and reload the latest data from the source files.
4. Save the file once refresh completes.

7. Publishing & Scheduled Refresh (for the report owner)

These steps are for whoever maintains the report, not for a general viewer:

1. In Power BI Desktop, go to Home → Publish → select the destination workspace.
2. Once published, open the report in the Power BI Service (app.powerbi.com).
3. Go to the associated dataset → Settings → 'Scheduled refresh'.
4. Turn Scheduled refresh ON, choose a frequency (e.g. daily), and set a time.
5. If the data source is a local file (not a cloud location like OneDrive/SharePoint), install and configure the On-premises Data Gateway so the Service can reach it — otherwise scheduled refresh will fail silently.
6. Save. The report will now update automatically on the schedule you set, without needing Power BI Desktop open.

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Total Listings	Availability Rate	Average Price	Scenario Price	Projected Revenue	Avg. Host Tenure (Years)
1749	53.53%	$109.92	$87.93	$629.35K	10.05

Timeline

26 December 2021 – 25 December 2022

Selected: 26 December 2021 – 25 December 2022

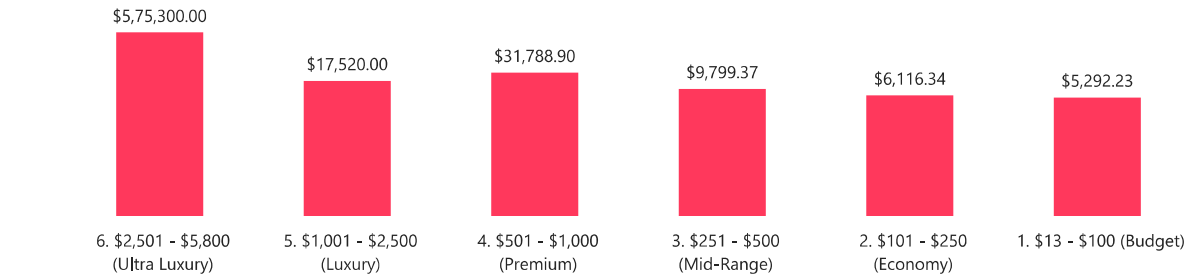
Room Type

- Deselect all
- Entire home/apt
- Hotel room
- Private room
- Shared room

Property Type

- Deselect all
- Boat
- Casa particular
- Castle
- Entire condominium (condo)

Average Revenue per Listing by Price Band



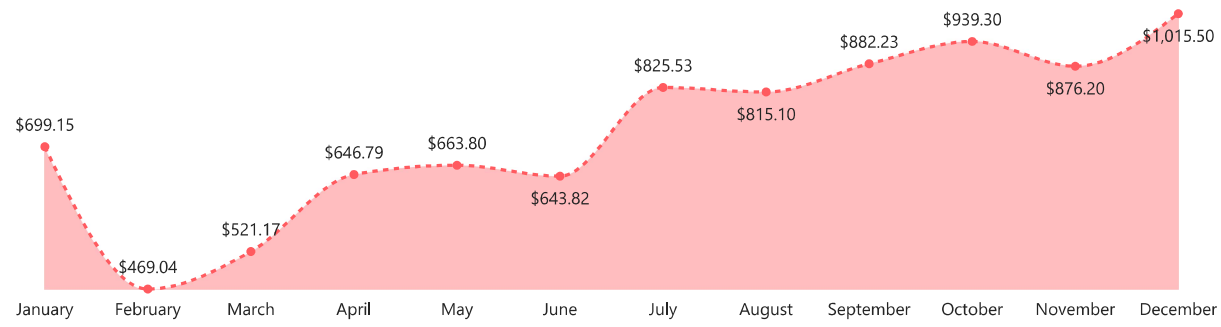
Occupancy Rate (What-If)

0.05

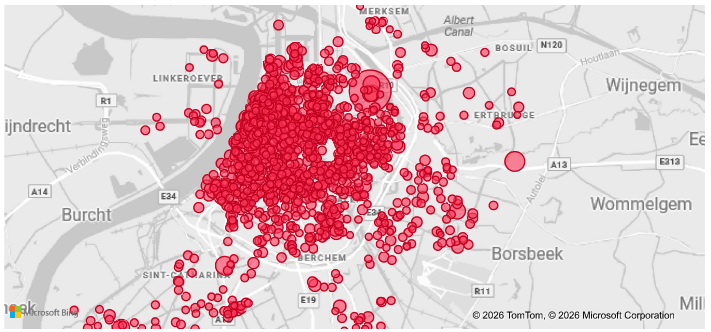
Seasonal Pricing Multiplier

0.80

Average Revenue per by Month

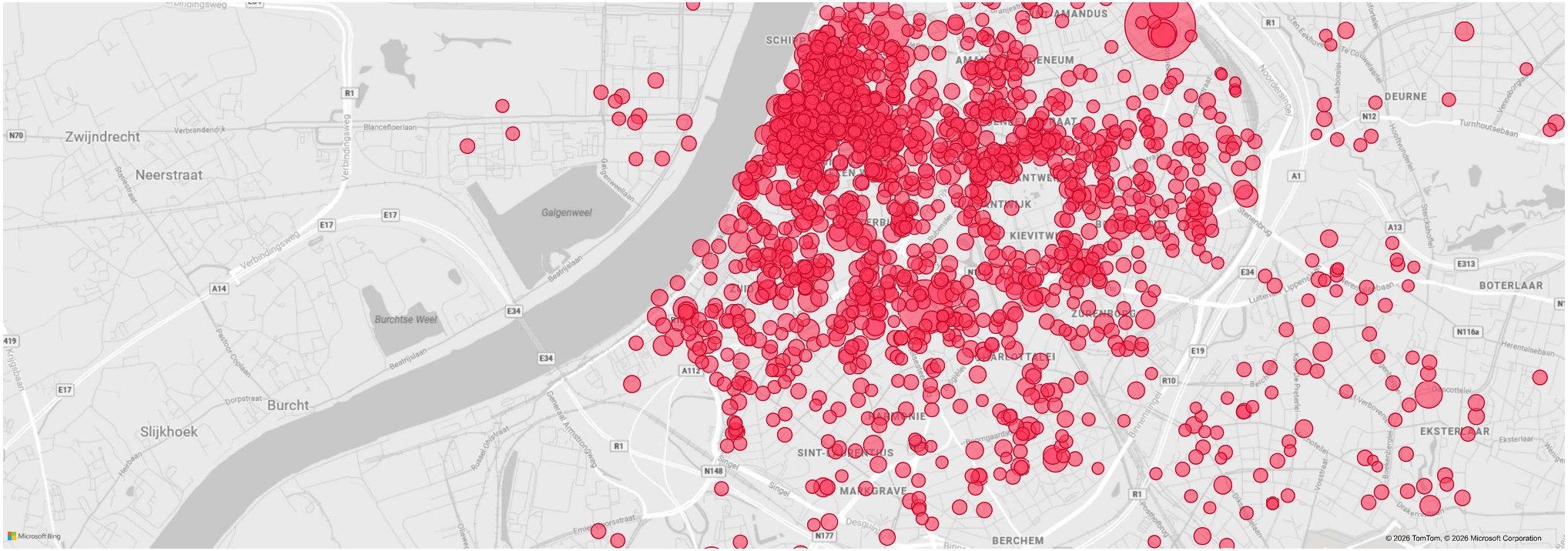


Antwerp, Belgium

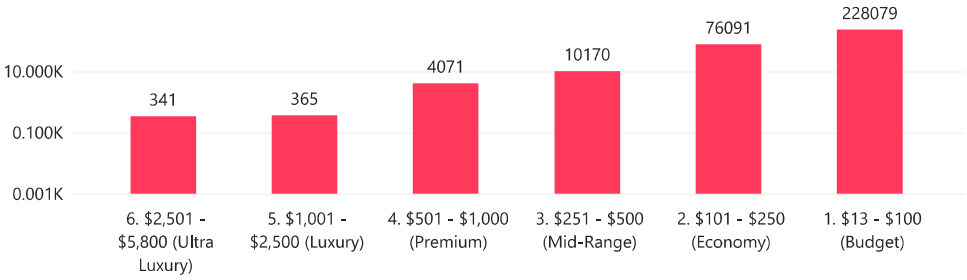


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Listings by Location



Count of Listed Hotels by Price Distribution



Top 20 Listings by Reviews

Property Name	Listing_id	Property_type	Average Price	Max of Beds	Review Count
Unique city loft with amazing view!	4502646	Entire loft	$78.89	2	648
Rustige studio	8766126	Entire rental unit	$60.00	1	564
Cosy apartment near Central Station	1901179	Entire rental unit	$64.00	4	463
Blue room_with private bathroom and kitchen	8587634	Entire residential home	$45.07	3	447
Total			**$67.73**	**4**	**7581**

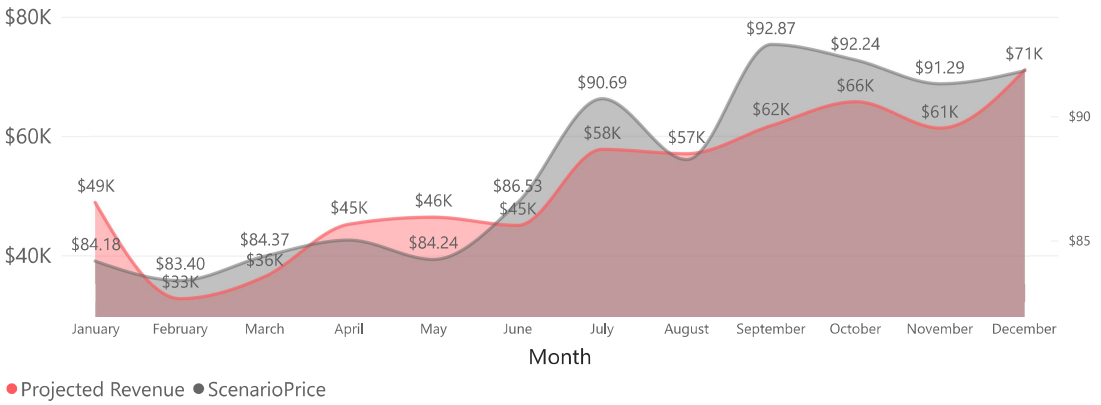
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Average Price	Scenario Price	Seasonal Pricing Multiplier	Projected Revenue	Occupancy Rate (What-If)
$109.92	$87.93	0.80	$629.35K	0.05

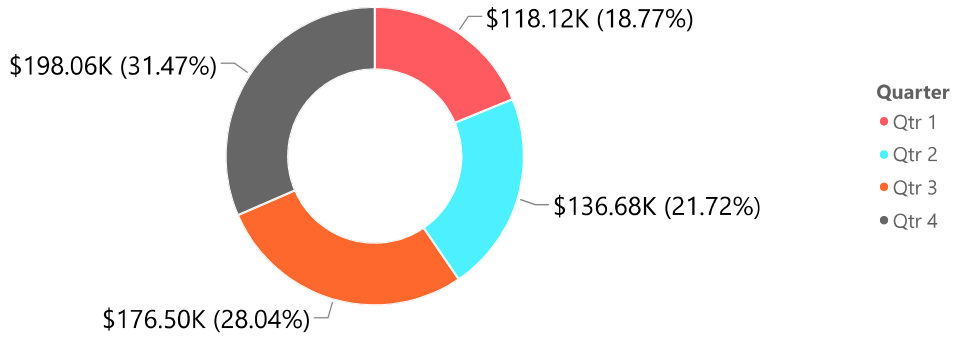
AveragePrice vs. ScenarioPrice



Projected Revenue and ScenarioPrice by Month



Projected Revenue by Quarter



Key Insights from the Dashboard

After analyzing the dashboard, I identified the following key insights.

1. Price Varies Significantly by Room Type: Hotel room (~$118.23/night) and Entire home/apt (~$114.71/night) command the highest prices, while Shared room (~$55.98/night, only 6 listings) sits far below. Hotel room pricing above full apartments suggests guests pay for a different kind of experience, not just space.

2. Reviews and Price Are Not Strongly Related: Correlation between average price and review count is near zero (-0.06). A well-reviewed listing cannot assume pricing power — price is driven more by room type, location, and amenities than guest satisfaction history.

3. Certain Amenities Carry a Real Price Premium, Others Do Not: Wifi and Heating show no premium — listings with them are actually priced lower, suggesting they're now baseline expectations. Air Conditioning (+$17.63/night) and Free Parking (+$12.41/night) carry genuine premiums and are worth highlighting instead.

4. Private Rooms in Townhouses Are the Most Efficient Earning Category: Best price-per-guest: Private room in townhouse (~$75.89/guest). Weakest: Entire residential home (~$26.60/guest) — more than double the gap, suggesting smaller shared-access properties are a stronger per-guest investment than large standalone rentals.

5. Listing Supply is Concentrated Among a Small Number of Hosts: The top 10 hosts control 12.5% of all 1,749 listings; one host alone owns 64.

6. Room Type Distribution Shapes Overall Market Character: Entire home/apt dominates by volume, Private room a distant second, Hotel/Shared room a small fraction.

Recommendations

1. **Test price increases on Entire home/apt listings** priced below the Hotel room benchmark.
2. **Shift Shared room listings to Private room** where possible — roughly 64% more revenue.
3. **Market A/C and parking, not Wifi/Heating** — the latter no longer differentiate pricing.
4. **Prioritize private-room-in-townhouse investment** for the strongest return per guest.
5. **Monitor multi-listing host concentration** given its regulatory implications.
6. **Don't use review count as a pricing signal** — price and reviews are unrelated here.



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Listing ID	Listing ID	Average Price	Reviews
4502646	Unique city loft with amazing view!	$78.89	648

Room Type

- Deselect all
- Entire home/apt

Host Details

Host_name	Host_Country	HostAgeYears	HasBio	Host_since	Host_location
Yara	Belgium	12.5	True	12-03-2014	Antwerp, Flanders, Belgium

Amenity Details

Amenity
Bathtub
Bed linens
Board games

Property Reviews

Date	Reviewer_name	comments
01-01-2015	Lou	The apartment is absolutely amazing : very wide, the greatest location ever, very white and clean and well equipped. Yara is an absolutely perfect host, very careful we'll come back here if we travel again to Antwerp!
01-02-2016	Frans	This place contributed to a great time in Antwerp. The apartment is comfy, spacious and clean. Its central location made it even more fun. There's tons of stuff to do in th of inspiration, Yara left soma notes on some cool places in the area that you could visit. On forehand of visiting I read a review that said the bed was pretty hard. Hc experience. The bed was pretty comfortable actually!
01-02-2017	Jane	Yara's place is just perfect! Clean and cozy, really good view over the window! Very bright and decoration is unique which I very appreciate! Everything in the apartment i location is great, very center. All the restaurants, the bar, the shops, the cafe just downstairs of your street. And you have a view of the beautiful church by your window.< about arranging everything. Thank you so much! The only thing you may need to know is no elevator, which is not a concern to me, but if your luggage was too he
01-03-2017	Kris	Yara, bedankt dat we hier een nachtje mochten slapen! Dit was onze eerste ervaring met Airbnb en het is ons heel goed bevallen! Het appartementje ligt heel centraal er de pizzeria die er naast ligt is erg lekker! Wellicht tot een volgende keer!
01-04-2019	Yohan	Placé au centre historique d'Anvers, cet appartement était superbe avec tous ce qu'il faut pour passé un super séjour, de plus Yara est une hôte très sympathique et très recommande les yeux fermé.

AveragePrice

$109.92

ReviewCount

62987

MonthlyRevenue

$15.73M

Revenue by Month

